AirTransformer

Cooling for urban critical facilities

A TRL6 machine
for heat driven cooling and water saving

Savery system IP and CAU materials

EIC Accelerator 2026 C05
Blended finance

Urban cooling under power and water constraints

Customer value at equal delivered cooling and stated operating conditions

60–70% lower electricity

Engineering objective
versus reference compression cooling

40–60% less external water

Qualified configurations
with a complete installation balance

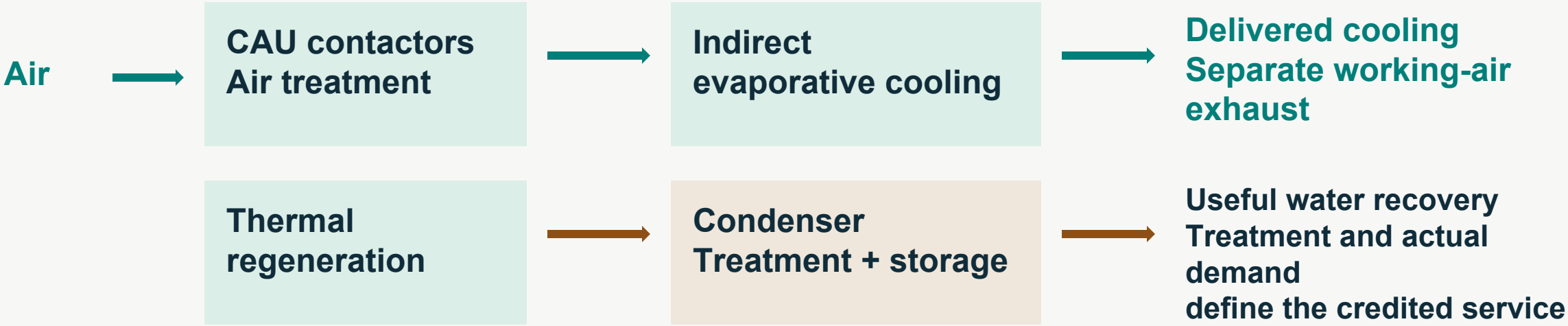
Owner approves investment. Engineer specifies. Operator accepts.

Continuity
Compliant hours and recovery under utility limits

Customer value
Net energy, water, installation and maintenance cost

Heat driven cooling with defined site interfaces

Product service architecture and qualification interfaces



Configuration and cycle timing follow the assigned duty

Independent safeguards with a controlled supervisory upgrade

Inputs: 50–70°C regeneration heat, compatible sink and auxiliary electricity

External water withdrawal includes internal use, evaporation, treatment and heat rejection.

Engineering service schematic; selected configuration determines supplied equipment

Savery IP supports a product engineering programme

Owned system IP, CAU processing and controlled product upgrades

01

Savery system IP

2019 priority family
EP3825627A1
WO2021099546A1

02

CAU process transfer

Shaping performance
Batch consistency
Manufacturing specifications

03

Control upgrade

Rules and safe fallback
MPC and supervised RL
Validated operating limits

Product authority retains process know-how, software and release control

Current TRL6 and a funded route to deployment

Existing machine, CETP field progression and EIC product qualification

TRL6

Current complete machine

Integrated product foundation

TRL7

CETP development target

Awarded grant, 2026–2029

TRL8

EIC product qualification

Planned by March 2029

HPE provides the CETP site and an established collaboration path.

EIC adds manufacture, reliability, compliance and repeatable deployment.

CETP and EIC reuse knowledge with separate work and cost allocation.

The EIC schedule has its own qualification route through M24.

Current machine and funded CETP status supplied by Savery; proposed EIC April 2027–March 2029

Competition is decided by the complete cooling service

Customer alternatives need a matched duty and whole installation comparison

Purchasing route	Disclosed offer / retained option
AirJoule / BitSink	Sorbent water, dry air and cooling
Transaera	MOF outdoor-air treatment
Mojave ArctiDry Hydro	Liquid-desiccant hydronic air handling
Ahbstra ARK	MOF atmospheric water harvesting
Munters SyCool	Waterless cooling with compressor support
Conventional retrofit	Compression, economisation and water reuse

Savery: air-side service, CAU contactors, process control and accountable delivery

Compare electricity, heat, external water, installation, maintenance and accepted cooling.

Manufacturer disclosures: AirJoule, Transaera, Mojave, Ahbstra and Munters

An initial €18m annual serviceable market in 2031

Urban colocation entry in Frankfurt, London, Amsterdam, Paris and Dublin

01	→ 02	→ 03	→ 04
600 MW	**10% air-side duty**	**25% joint fit**	**€1,200 per kW**
Annual capacity flow Flat planning reference	60 MW assigned cooling requirement	15 MW meeting heat, sink, climate and site needs	Equipment, installation and application engineering

Management planning case; €7.2–28.8m at 10–40% joint qualification.

2031 C05 sales target €1.8m gives 10% share in the same scope.

CBRE European Data Centres Q3 2025 capacity reference; Savery assumptions and pricing scenario

A staged entry plan links sales to qualified capacity

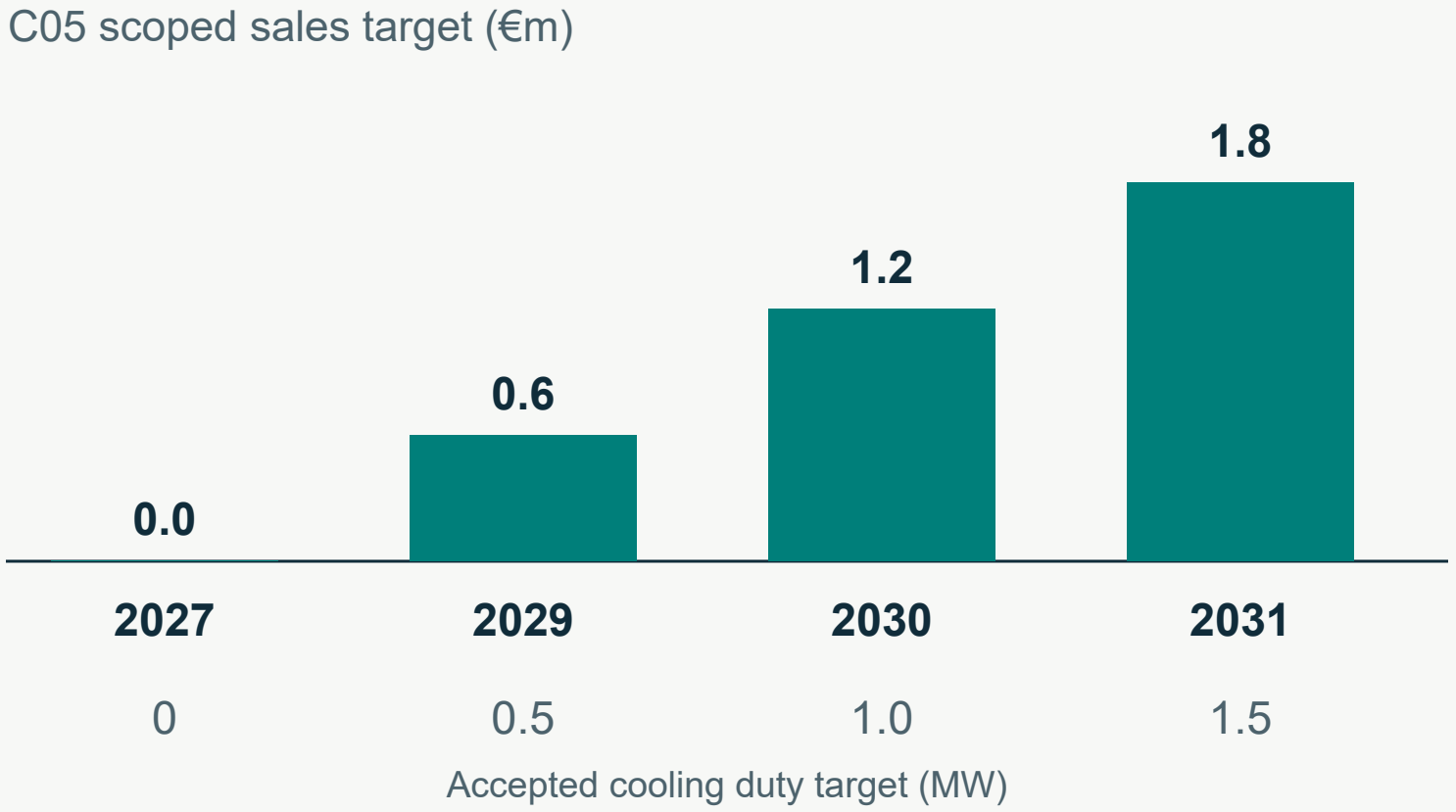
Initial C05 scope; equipment, installation and engineering revenue

€120,000

100 kW planning equivalent

€110,000 hardware
€8,000 installation
€2,000 engineering

Configuration-specific offers follow product qualification. Maintenance is separate.



Savery C05 market-entry scenario; qualified manufacturing and service capacity constrain sales

European manufacturing and release capability

Critical processing, assembly, calibration and lifecycle feedback

Process

Coating, bonding and sealing

Engineering baseline

Requirements and interfaces

M6 design release

Trace

Batch, model and software versions

Manufacturing transfer

Six qualification batches

M18 batch release

Release

Cyclic capacity, leakage and interfaces

Product qualification

Reliability and conformity

M24 TRL8 dossier

Improve

Field evidence, repair and requalification

Commercial deployment

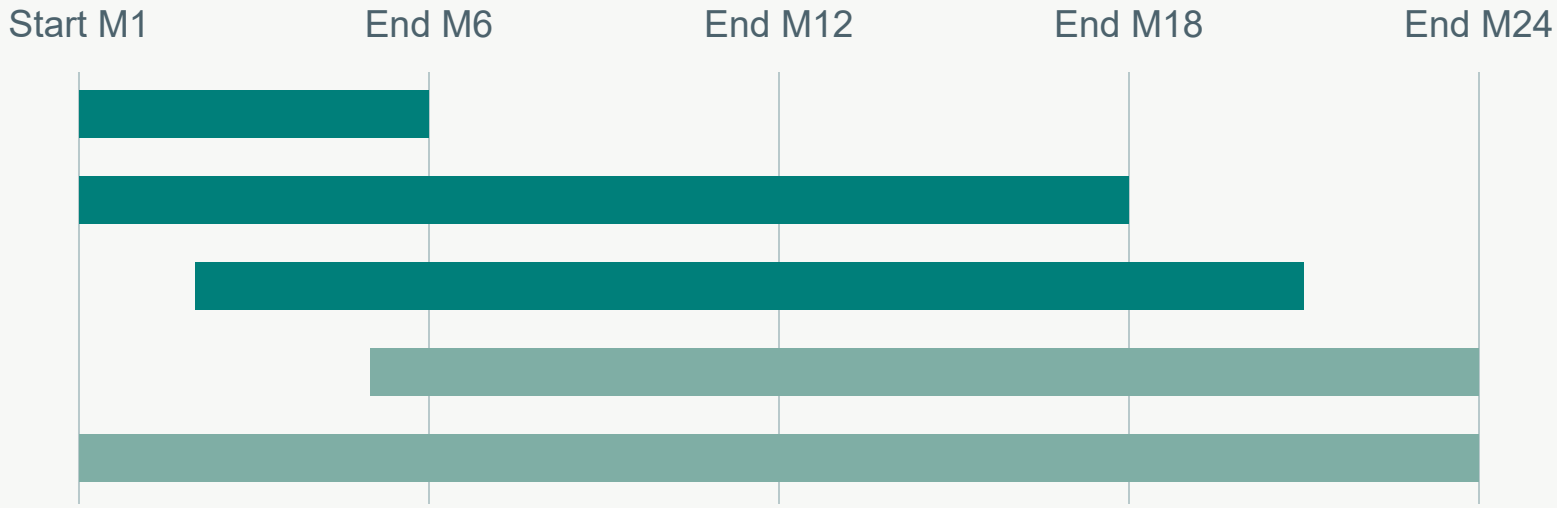
Small-batch capacity ramp

2029–2031 sales gates

Savery owns its system IP; product and territory FTO review precedes commercial commitments.

A resourced 24 month product qualification action

258 person-months across five grant work packages



M6
Engineering baseline

M12
Manufacturing design

M21
Conformity dossier

M24
TRL8 qualification

M18 batch and reliability gate; investment work separately prepares production and deployment.

A core team with scientific and execution depth

Named operational ownership, scientific board and budgeted specialist roles

Frédéric Cui	System architecture and product decisions
Qin Yu	Engineering, suppliers and commissioning
Ina Wang	Customer requirements and acceptance
Norbert Stock	Scientific board, CAU materials
Aline Auroux	Scientific board, adsorption and thermodynamics

258 PM

14 planned execution roles

Modelling and controls
Process, quality and reliability
Testing, field and software

Gender measures cover the whole organisation.

Core project commitments: Frédéric 50%, Qin 87.5%, Ina 50%; board contribution separate from paid PM

Blended finance for productisation and deployment

Grant and investment have distinct uses, agreements and payment processes

Eligible action costs: €3,550,833.33

Grant €2,485,583.33 | **Non-grant €1.065m**

Seed target: €8m

EIC investment €2m | **Other sources €6m target**

€1.9m investment for deployment and €0.1m for innovation co-financing.

Total EIC request €4,485,583.33

70% grant financing; non-grant share €1,065,250; investment forms part of the €8m Seed target